



BEST PRACTICE SERIES

Wheel Loader Lubricated Pin Inspection

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
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1.0 Introduction

The purpose of this best practice is to establish a preventive maintenance procedure to avoid unexpected failures of grease lubricated pin joints in Large Wheel Loaders. Depending on the equipment application, this procedure will allow for wear trending and aid in determining useful life of the pin and successfully completing repairs before a failure occurs.

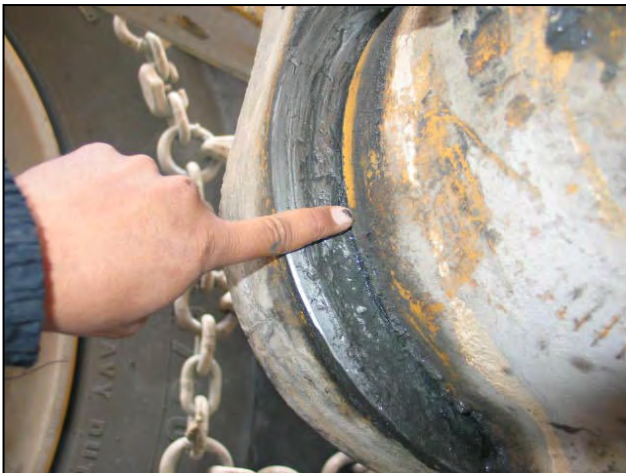
By inspecting greased lubricated pins periodically, we can:

- Identify a temperature increase related to the wear of the pin or failure on the lubrication system.
- Optimize grease use on every pin.

Grease lubricated pin joints can have sudden failures, which cause issues with the production and maintenance flow of a mine site. Sometimes more grease than is needed is used, contaminating the environment and increasing operation costs. This best practice will aid in reducing this occurrence.

The pin inspection process and subsequent data analysis will aid in avoiding unscheduled downtime and help to increase machine availability.

By optimizing the amount of grease used we will reduce Owning and Operating costs and also avoid contamination of the environment.



These pictures show the effects of uncontrolled wear and lubrication problems.

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Wheel Loader Grease Lubricated Pin Inspection

DATE
6/17/2011

CHG
NO
01

NUMBER
0909-1.01-1183

2.0 Best Practice Description

The full pin inspection includes:

- Measuring the joint temperature immediately after operation, at refueling or shift changes, and during Pre-PM inspections.
- Measuring the amount of grease injected into each pin.
- Document the temperatures and volume of grease injected.

In order to perform the best practice an infrared thermometer will be needed to read pin joint temperatures. To measure the injected grease volume, the mechanic will need a measured container.

The information must be recorded, so that the data can be trended and analyzed to help achieve accurate repair before failure.

As pin operating temperatures will change due to ambient temperatures and operating conditions different baselines may need to be set for each machine. The variation from these baselines will indicate that a failure is immanent.



3.0 Implementation Steps

1. Inform the preventive maintenance manager and inspectors of the process
2. Establish who will be responsible for carrying out the procedure
3. Establish baseline pin temperatures
4. Begin regularly scheduled data collection.

In order to set a baseline pin temperature a machine in good condition should be operated in its normal operation environment. The various pin temperatures should then be recorded along with the ambient temperature. The delta between the pin temperature and the ambient temperature will be monitored for deviation from the baseline.

Once the process has been implemented and a baseline set if during future inspections deviation is noted between the baseline and the actual temperature reading a potential problem may be indicated.

This best practice has been set up in various mine sites over the past 5 years where the implementation process took less than a month. The areas involved were Planning and Maintenance.

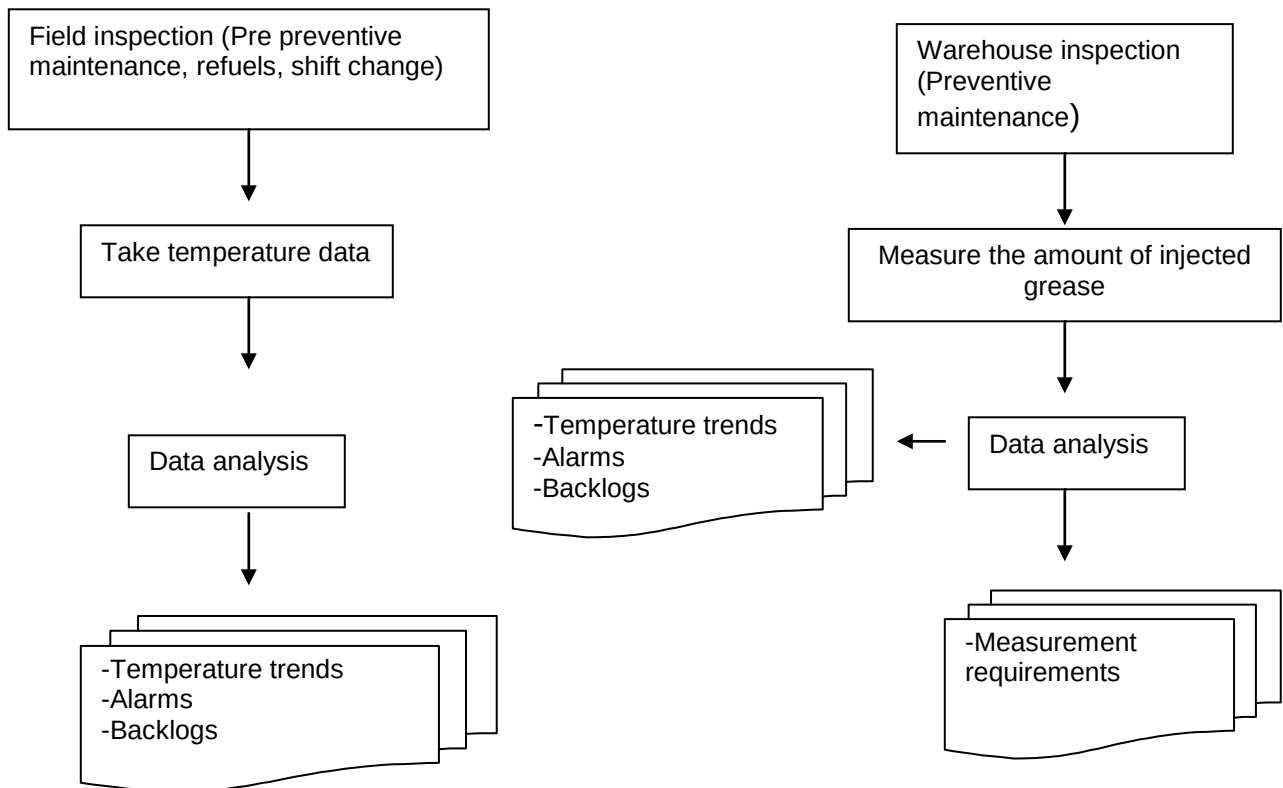
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4.0 Benefits

Implementation of this best practice can aid in reducing costly pin joint failures and the unscheduled downtime associated with these failures.

Experience has shown that a specific 994F wheel loader with no inspections of the pins or grease system presented major joint failures. In this instance the repair required two D pins (\$3699USD) and an A pin (\$2,499USD), these costs do not include labor to repair or downtime. The failures occurred during the first 4000 hours of operation.

Another machine, which experiences a major load, had experienced 3 pin D failures in the first 2000 operation hours (\$5,400USD)

Loaders placed into service after the implementation of this best practice in 2007 have had no failures related to the pin joints.

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5.0 Resources Required

- Field inspector
- Infrared thermometer
- Measured containers
- Maintenance personnel
- 4 hour training per person

6.0 Supporting Attachments / References

7.0 Related Best Practices

8.0 Acknowledgements

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